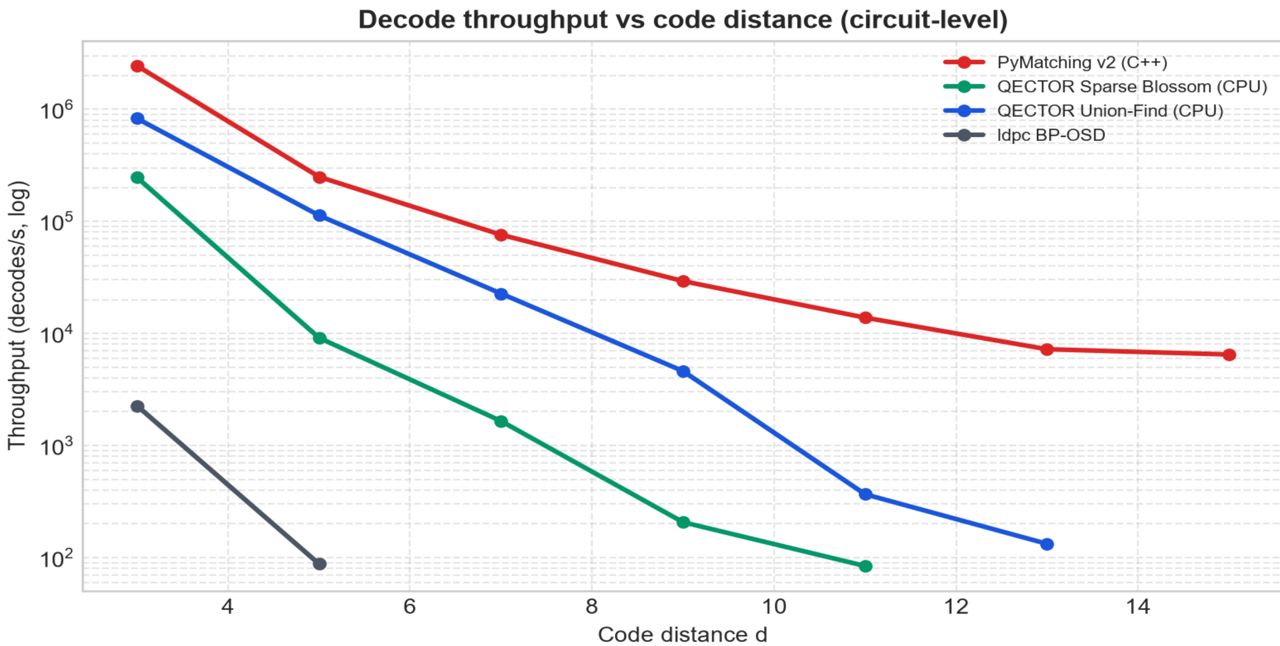
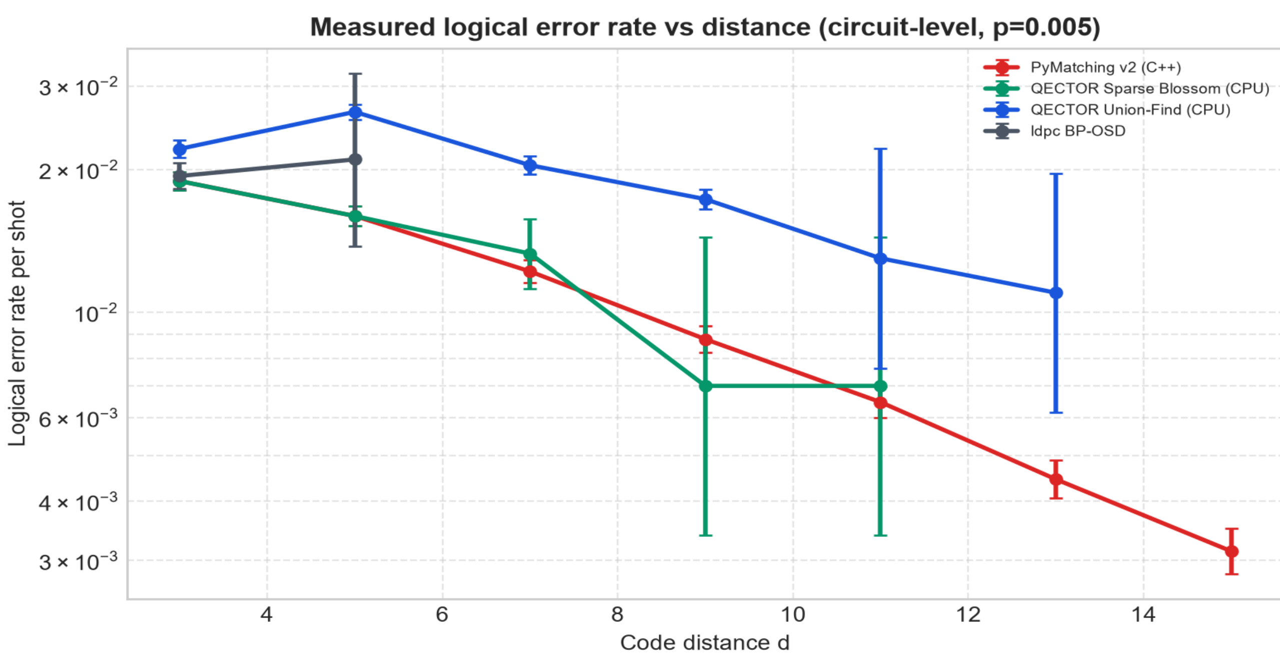
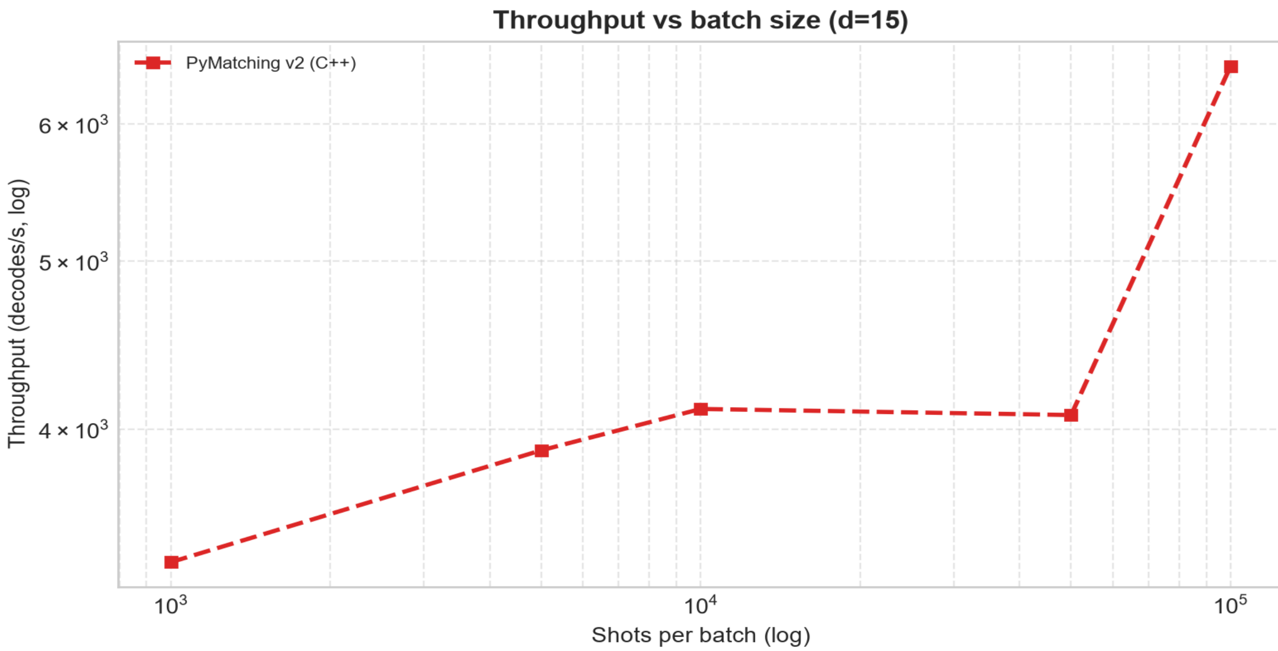
# QECTOR v0.7.0 — circuit-level decoder comparison

Measured with ler.estimate_ler_circuit_level: identical circuit, DEM, samples and decode_batch resolver for every decoder, scored against the circuit's logical observables. Validated by ler.assert_comparable. No extrapolated cells.

**Noise model:** circuit-level, p=0.005 | **Commit:** b0f5456470 | **Generated:** 2026-07-31T23:03:23Z
**Platform:** Windows-10-10.0.26100-SP0 | **Python:** 3.11.9




Throughput vs batch size (d=15)
PyMatching v2 (C++)
Throughput (decodes/s, log)
6 × 10^3
5 × 10^3
4 × 10^3
10^3
10^4
10^5
Shots per batch (log)

## Measured results (77 cells)

| Decoder | d | Shots | Errors | LER | 95% CI | dec/s | vs PM |
|---|---|---|---|---|---|---|---|
| PyMatching v2 (C++) | 3 | 1,000 | 13 | **0.01300** | [0.0076, 0.0221] | **1,949,318** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 3 | 1,000 | 13 | **0.01300** | [0.0076, 0.0221] | **154,269** | 0.08x |
| QECTOR Union-Find (CPU) | 3 | 1,000 | 16 | **0.01600** | [0.0099, 0.0258] | **222,321** | 0.11x |
| ldpc BP-OSD | 3 | 1,000 | 12 | **0.01200** | [0.0069, 0.0209] | **2,267** | 0.00x |
| PyMatching v2 (C++) | 3 | 5,000 | 84 | **0.01680** | [0.0136, 0.0208] | **1,865,811** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 3 | 5,000 | 84 | **0.01680** | [0.0136, 0.0208] | **236,337** | 0.13x |
| QECTOR Union-Find (CPU) | 3 | 5,000 | 105 | **0.02100** | [0.0174, 0.0254] | **481,024** | 0.26x |
| ldpc BP-OSD | 3 | 5,000 | 87 | **0.01740** | [0.0141, 0.0214] | **2,286** | 0.00x |
| PyMatching v2 (C++) | 3 | 10,000 | 202 | **0.02020** | [0.0176, 0.0231] | **1,848,941** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 3 | 10,000 | 202 | **0.02020** | [0.0176, 0.0231] | **248,616** | 0.13x |
| QECTOR Union-Find (CPU) | 3 | 10,000 | 233 | **0.02330** | [0.0205, 0.0264] | **571,932** | 0.31x |
| ldpc BP-OSD | 3 | 10,000 | 199 | **0.01990** | [0.0173, 0.0228] | **2,330** | 0.00x |
| PyMatching v2 (C++) | 3 | 50,000 | 960 | **0.01920** | [0.0180, 0.0204] | **2,234,837** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 3 | 50,000 | 960 | **0.01920** | [0.0180, 0.0204] | **250,396** | 0.11x |
| QECTOR Union-Find (CPU) | 3 | 50,000 | 1128 | **0.02256** | [0.0213, 0.0239] | **812,806** | 0.36x |
| ldpc BP-OSD | 3 | 50,000 | 969 | **0.01938** | [0.0182, 0.0206] | **2,259** | 0.00x |
| PyMatching v2 (C++) | 3 | 100,000 | 1891 | **0.01891** | [0.0181, 0.0198] | **2,437,651** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 3 | 100,000 | 1891 | **0.01891** | [0.0181, 0.0198] | **245,629** | 0.10x |
| QECTOR Union-Find (CPU) | 3 | 100,000 | 2210 | **0.02210** | [0.0212, 0.0230] | **825,726** | 0.34x |
| PyMatching v2 (C++) | 5 | 1,000 | 19 | **0.01900** | [0.0122, 0.0295] | **266,411** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 5 | 1,000 | 19 | **0.01900** | [0.0122, 0.0295] | **8,690** | 0.03x |
| QECTOR Union-Find (CPU) | 5 | 1,000 | 26 | **0.02600** | [0.0178, 0.0378] | **79,888** | 0.30x |
| ldpc BP-OSD | 5 | 1,000 | 21 | **0.02100** | [0.0138, 0.0319] | **88** | — |
| PyMatching v2 (C++) | 5 | 5,000 | 73 | **0.01460** | [0.0116, 0.0183] | **286,110** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 5 | 5,000 | 73 | **0.01460** | [0.0116, 0.0183] | **9,018** | 0.03x |
| QECTOR Union-Find (CPU) | 5 | 5,000 | 148 | **0.02960** | [0.0253, 0.0347] | **91,363** | 0.32x |
| PyMatching v2 (C++) | 5 | 10,000 | 166 | **0.01660** | [0.0143, 0.0193] | **234,845** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 5 | 10,000 | 166 | **0.01660** | [0.0143, 0.0193] | **9,204** | 0.04x |
| QECTOR Union-Find (CPU) | 5 | 10,000 | 268 | **0.02680** | [0.0238, 0.0302] | **97,208** | 0.41x |
| PyMatching v2 (C++) | 5 | 50,000 | 809 | **0.01618** | [0.0151, 0.0173] | **225,455** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 5 | 50,000 | 810 | **0.01620** | [0.0151, 0.0173] | **8,990** | 0.04x |
| QECTOR Union-Find (CPU) | 5 | 50,000 | 1356 | **0.02712** | [0.0257, 0.0286] | **113,116** | 0.50x |
| PyMatching v2 (C++) | 5 | 100,000 | 1596 | **0.01596** | [0.0152, 0.0168] | **249,065** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 5 | 100,000 | 1596 | **0.01596** | [0.0152, 0.0168] | **9,121** | 0.04x |
| QECTOR Union-Find (CPU) | 5 | 100,000 | 2645 | **0.02645** | [0.0255, 0.0275] | **112,797** | 0.45x |
| PyMatching v2 (C++) | 7 | 1,000 | 11 | **0.01100** | [0.0062, 0.0196] | **54,523** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 7 | 1,000 | 11 | **0.01100** | [0.0062, 0.0196] | **1,631** | 0.03x |
| QECTOR Union-Find (CPU) | 7 | 1,000 | 18 | **0.01800** | [0.0114, 0.0283] | **9,142** | 0.17x |
| PyMatching v2 (C++) | 7 | 5,000 | 74 | **0.01480** | [0.0118, 0.0185] | **62,572** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 7 | 5,000 | 75 | **0.01500** | [0.0120, 0.0188] | **1,565** | 0.03x |
| QECTOR Union-Find (CPU) | 7 | 5,000 | 107 | **0.02140** | [0.0177, 0.0258] | **20,731** | 0.33x |
| PyMatching v2 (C++) | 7 | 10,000 | 130 | **0.01300** | [0.0110, 0.0154] | **89,104** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 7 | 10,000 | 133 | **0.01330** | [0.0112, 0.0157] | **1,651** | 0.02x |
| QECTOR Union-Find (CPU) | 7 | 10,000 | 209 | **0.02090** | [0.0183, 0.0239] | **22,427** | 0.25x |

| Decoder | d | Shots | Errors | LER | 95% CI | dec/s | vs PM |
|---|---|---|---|---|---|---|---|
| PyMatching v2 (C++) | 7 | 50,000 | 619 | **0.01238** | [0.0114, 0.0134] | **69,244** | 1.00x |
| QECTOR Union-Find (CPU) | 7 | 50,000 | 1055 | **0.02110** | [0.0199, 0.0224] | **21,552** | 0.31x |
| PyMatching v2 (C++) | 7 | 100,000 | 1220 | **0.01220** | [0.0115, 0.0129] | **75,767** | 1.00x |
| QECTOR Union-Find (CPU) | 7 | 100,000 | 2042 | **0.02042** | [0.0196, 0.0213] | **22,643** | 0.30x |
| PyMatching v2 (C++) | 9 | 1,000 | 7 | **0.00700** | [0.0034, 0.0144] | **16,122** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 9 | 1,000 | 7 | **0.00700** | [0.0034, 0.0144] | **207** | 0.01x |
| QECTOR Union-Find (CPU) | 9 | 1,000 | 14 | **0.01400** | [0.0084, 0.0234] | **1,404** | 0.09x |
| PyMatching v2 (C++) | 9 | 5,000 | 36 | **0.00720** | [0.0052, 0.0100] | **28,259** | 1.00x |
| QECTOR Union-Find (CPU) | 9 | 5,000 | 85 | **0.01700** | [0.0138, 0.0210] | **4,692** | 0.17x |
| PyMatching v2 (C++) | 9 | 10,000 | 78 | **0.00780** | [0.0063, 0.0097] | **27,630** | 1.00x |
| QECTOR Union-Find (CPU) | 9 | 10,000 | 156 | **0.01560** | [0.0134, 0.0182] | **3,812** | 0.14x |
| PyMatching v2 (C++) | 9 | 50,000 | 450 | **0.00900** | [0.0082, 0.0099] | **29,152** | 1.00x |
| QECTOR Union-Find (CPU) | 9 | 50,000 | 827 | **0.01654** | [0.0155, 0.0177] | **4,599** | 0.16x |
| PyMatching v2 (C++) | 9 | 100,000 | 878 | **0.00878** | [0.0082, 0.0094] | **29,244** | 1.00x |
| QECTOR Union-Find (CPU) | 9 | 100,000 | 1732 | **0.01732** | [0.0165, 0.0181] | **4,606** | 0.16x |
| PyMatching v2 (C++) | 11 | 1,000 | 7 | **0.00700** | [0.0034, 0.0144] | **11,490** | 1.00x |
| QECTOR Sparse Blossom (CPU) | 11 | 1,000 | 7 | **0.00700** | [0.0034, 0.0144] | **84** | 0.01x |
| QECTOR Union-Find (CPU) | 11 | 1,000 | 13 | **0.01300** | [0.0076, 0.0221] | **367** | 0.03x |
| PyMatching v2 (C++) | 11 | 5,000 | 24 | **0.00480** | [0.0032, 0.0071] | **12,517** | 1.00x |
| PyMatching v2 (C++) | 11 | 10,000 | 77 | **0.00770** | [0.0062, 0.0096] | **13,693** | 1.00x |
| PyMatching v2 (C++) | 11 | 50,000 | 334 | **0.00668** | [0.0060, 0.0074] | **14,040** | 1.00x |
| PyMatching v2 (C++) | 11 | 100,000 | 647 | **0.00647** | [0.0060, 0.0070] | **13,816** | 1.00x |
| PyMatching v2 (C++) | 13 | 1,000 | 3 | **0.00300** | [0.0010, 0.0088] | **7,384** | 1.00x |
| QECTOR Union-Find (CPU) | 13 | 1,000 | 11 | **0.01100** | [0.0062, 0.0196] | **133** | 0.02x |
| PyMatching v2 (C++) | 13 | 5,000 | 18 | **0.00360** | [0.0023, 0.0057] | **7,576** | 1.00x |
| PyMatching v2 (C++) | 13 | 10,000 | 37 | **0.00370** | [0.0027, 0.0051] | **6,933** | 1.00x |
| PyMatching v2 (C++) | 13 | 50,000 | 221 | **0.00442** | [0.0039, 0.0050] | **7,243** | 1.00x |
| PyMatching v2 (C++) | 13 | 100,000 | 445 | **0.00445** | [0.0041, 0.0049] | **7,218** | 1.00x |
| PyMatching v2 (C++) | 15 | 1,000 | 2 | **0.00200** | [0.0005, 0.0073] | **3,351** | 1.00x |
| PyMatching v2 (C++) | 15 | 5,000 | 12 | **0.00240** | [0.0014, 0.0042] | **3,888** | 1.00x |
| PyMatching v2 (C++) | 15 | 10,000 | 44 | **0.00440** | [0.0033, 0.0059] | **4,108** | 1.00x |
| PyMatching v2 (C++) | 15 | 50,000 | 171 | **0.00342** | [0.0029, 0.0040] | **4,075** | 1.00x |
| PyMatching v2 (C++) | 15 | 100,000 | 314 | **0.00314** | [0.0028, 0.0035] | **6,480** | 1.00x |

## Not measured (63 cells)

Listed so the gaps are explicit. These cells were never run and carry no numbers; nothing here was extrapolated or synthesised.

| Decoder | d | Shots | Reason | Projected |
|---|---|---|---|---|
| ldpc BP-OSD | 3 | 100,000 | over per-cell decode budget | 41s |
| ldpc BP-OSD | 5 | 5,000 | over per-cell decode budget | 61s |
| ldpc BP-OSD | 5 | 10,000 | over per-cell decode budget | 122s |
| ldpc BP-OSD | 5 | 50,000 | over per-cell decode budget | 609s |
| ldpc BP-OSD | 5 | 100,000 | over per-cell decode budget | 1,217s |
| QECTOR Sparse Blossom (CPU) | 7 | 50,000 | over per-cell decode budget | 36s |
| QECTOR Sparse Blossom (CPU) | 7 | 100,000 | over per-cell decode budget | 71s |
| ldpc BP-OSD | 7 | 1,000 | over per-cell decode budget | 53s |
| ldpc BP-OSD | 7 | 5,000 | over per-cell decode budget | 263s |
| ldpc BP-OSD | 7 | 10,000 | over per-cell decode budget | 526s |
| ldpc BP-OSD | 7 | 50,000 | over per-cell decode budget | 2,629s |
| ldpc BP-OSD | 7 | 100,000 | over per-cell decode budget | 5,258s |
| QECTOR Sparse Blossom (CPU) | 9 | 5,000 | over per-cell decode budget | 26s |
| QECTOR Sparse Blossom (CPU) | 9 | 10,000 | over per-cell decode budget | 53s |
| QECTOR Sparse Blossom (CPU) | 9 | 50,000 | over per-cell decode budget | 263s |
| QECTOR Sparse Blossom (CPU) | 9 | 100,000 | over per-cell decode budget | 525s |
| ldpc BP-OSD | 9 | 1,000 | over per-cell decode budget | 119s |
| ldpc BP-OSD | 9 | 5,000 | over per-cell decode budget | 597s |
| ldpc BP-OSD | 9 | 10,000 | over per-cell decode budget | 1,193s |
| ldpc BP-OSD | 9 | 50,000 | over per-cell decode budget | 5,967s |
| ldpc BP-OSD | 9 | 100,000 | over per-cell decode budget | 11,935s |
| QECTOR Sparse Blossom (CPU) | 11 | 5,000 | over per-cell decode budget | 61s |
| QECTOR Sparse Blossom (CPU) | 11 | 10,000 | over per-cell decode budget | 122s |
| QECTOR Sparse Blossom (CPU) | 11 | 50,000 | over per-cell decode budget | 609s |
| QECTOR Sparse Blossom (CPU) | 11 | 100,000 | over per-cell decode budget | 1,219s |
| QECTOR Union-Find (CPU) | 11 | 5,000 | over per-cell decode budget | 29s |
| QECTOR Union-Find (CPU) | 11 | 10,000 | over per-cell decode budget | 58s |
| QECTOR Union-Find (CPU) | 11 | 50,000 | over per-cell decode budget | 291s |
| QECTOR Union-Find (CPU) | 11 | 100,000 | over per-cell decode budget | 581s |
| ldpc BP-OSD | 11 | 1,000 | over per-cell decode budget | 271s |
| ldpc BP-OSD | 11 | 5,000 | over per-cell decode budget | 1,353s |
| ldpc BP-OSD | 11 | 10,000 | over per-cell decode budget | 2,706s |
| ldpc BP-OSD | 11 | 50,000 | over per-cell decode budget | 13,530s |
| ldpc BP-OSD | 11 | 100,000 | over per-cell decode budget | 27,059s |
| QECTOR Sparse Blossom (CPU) | 13 | 1,000 | over per-cell decode budget | 39s |
| QECTOR Sparse Blossom (CPU) | 13 | 5,000 | over per-cell decode budget | 194s |
| QECTOR Sparse Blossom (CPU) | 13 | 10,000 | over per-cell decode budget | 387s |
| QECTOR Sparse Blossom (CPU) | 13 | 50,000 | over per-cell decode budget | 1,937s |
| QECTOR Sparse Blossom (CPU) | 13 | 100,000 | over per-cell decode budget | 3,875s |
| QECTOR Union-Find (CPU) | 13 | 5,000 | over per-cell decode budget | 75s |
| QECTOR Union-Find (CPU) | 13 | 10,000 | over per-cell decode budget | 149s |
| QECTOR Union-Find (CPU) | 13 | 50,000 | over per-cell decode budget | 746s |
| QECTOR Union-Find (CPU) | 13 | 100,000 | over per-cell decode budget | 1,492s |

| Decoder | d | Shots | Reason | Projected |
|---|---|---|---|---|
| ldpc BP-OSD | 13 | 1,000 | over per-cell decode budget | 452s |
| ldpc BP-OSD | 13 | 5,000 | over per-cell decode budget | 2,258s |
| ldpc BP-OSD | 13 | 10,000 | over per-cell decode budget | 4,516s |
| ldpc BP-OSD | 13 | 50,000 | over per-cell decode budget | 22,578s |
| ldpc BP-OSD | 13 | 100,000 | over per-cell decode budget | 45,156s |
| QECTOR Sparse Blossom (CPU) | 15 | 1,000 | over per-cell decode budget | 95s |
| QECTOR Sparse Blossom (CPU) | 15 | 5,000 | over per-cell decode budget | 474s |
| QECTOR Sparse Blossom (CPU) | 15 | 10,000 | over per-cell decode budget | 949s |
| QECTOR Sparse Blossom (CPU) | 15 | 50,000 | over per-cell decode budget | 4,745s |
| QECTOR Sparse Blossom (CPU) | 15 | 100,000 | over per-cell decode budget | 9,489s |
| QECTOR Union-Find (CPU) | 15 | 1,000 | over per-cell decode budget | 43s |
| QECTOR Union-Find (CPU) | 15 | 5,000 | over per-cell decode budget | 213s |
| QECTOR Union-Find (CPU) | 15 | 10,000 | over per-cell decode budget | 426s |
| QECTOR Union-Find (CPU) | 15 | 50,000 | over per-cell decode budget | 2,132s |
| QECTOR Union-Find (CPU) | 15 | 100,000 | over per-cell decode budget | 4,264s |
| ldpc BP-OSD | 15 | 1,000 | over per-cell decode budget | 729s |
| ldpc BP-OSD | 15 | 5,000 | over per-cell decode budget | 3,647s |
| ldpc BP-OSD | 15 | 10,000 | over per-cell decode budget | 7,294s |
| ldpc BP-OSD | 15 | 50,000 | over per-cell decode budget | 36,468s |
| ldpc BP-OSD | 15 | 100,000 | over per-cell decode budget | 72,937s |